

The emerging role of glucagon-like peptide-1 receptor agonists in idiopathic intracranial hypertension: Therapeutic potential and challenges

Mangayakarasi Thandampallayam, Khawla Abusamra¹, Ashwini Kini², Rani P Vasireddy³

Weight reduction is a well-established disease-modifying intervention in the management of idiopathic intracranial hypertension (IIH), with clinical guidelines commonly recommending a target of at least 10% total body weight loss to achieve meaningful symptom improvement and reduction in intracranial pressure (ICP). Multiple modalities can be employed to facilitate weight loss, including lifestyle modification, pharmacologic therapy, and bariatric surgery. Among pharmacologic interventions, glucagon-like peptide-1 (GLP-1) receptor agonists (RAs) have demonstrated superior efficacy in achieving significant weight loss compared to conventional weight management strategies. Recent reports from ongoing studies have also indicated that GLP-1 RAs may reduce the need for acetazolamide, a standard treatment used to decrease cerebrospinal fluid (CSF) production in IIH, as they not only reduce ICP through weight loss but also have direct effects on CSF production. This review evaluates the role of GLP-1 RAs in IIH, highlighting their mechanisms, benefits, and emerging research. Challenges include limited IIH-specific data, high costs, side effects, and the need for long-term therapy. While early findings are promising, larger trials are needed. GLP-1 RAs should complement, not replace, comprehensive lifestyle and multidisciplinary IIH management strategies.

Key words: Glucagon-like peptide-1 receptor agonists, idiopathic intracranial hypertension, obesity, weight loss

Idiopathic intracranial hypertension (IIH) is characterized by raised intracranial pressure (ICP) without any identifiable cause. It is most commonly seen in young obese women of reproductive age. Common symptoms include headaches, tinnitus, and visual disturbances. Vision loss is the most serious complication, resulting from prolonged optic disc edema that can cause permanent damage. If untreated, patients may suffer from visual field loss, reduced visual acuity, or even permanent blindness, severely impacting their independence and quality of life. Weight gain is a major risk factor for the development of this condition. Weight loss of about 10% has shown to modify the disease process and if maintained can help induce remission. Hence, body weight is an important disease modifying factor associated with IIH and weight loss interventions can be an effective approach in modifying the course of IIH. The strong link between obesity and IIH has prompted investigation into whether glucagon-like peptide-1 (GLP-1) receptor agonists (RAs) could also be effective for managing IIH. Initial studies have shown encouraging results, suggesting that these drugs may offer a new and potentially more sustainable treatment approach.

There is a need for targeted therapy as there is an increase in the incidence of obesity globally. Acetazolamide is the most commonly used off-label medication for IIH. Other medications

like topiramate, furosemide, and spironolactone have also been used. Owing to side effects and sometimes treatment failures, new therapeutic agents like GLP-1 RAs have been studied in the management of IIH.^[1]

Methods

This narrative review examines the published literature on the use of GLP-1 RAs in individuals with IIH. A literature search was conducted in PubMed and supplemented by manual review of bibliographies from key publications. The search strategy used the keywords and Medical Subject Headings (MeSH) terms “idiopathic intracranial hypertension,” “pseudotumor cerebri,” “GLP-1 receptor agonist,” “semaglutide,” “dulaglutide,” and “tirzepatide” with Boolean operators (“AND,” “OR”) to refine results. The search covered publications from January 2000 through June 2024.

Eligible studies included human subjects with a confirmed diagnosis of IIH and encompassed clinical trials, retrospective cohort studies, case reports, and review articles assessing the use of GLP-1 RAs in IIH. Only publications available in English were included. Exclusion criteria were preclinical or animal studies (except those providing mechanistic insights), abstracts without accompanying published manuscripts, and articles

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Departments of Neurology and Ophthalmology, ³Neurology, University of Kentucky, Lexington, Kentucky, ¹Department of Neurology, Newton Medical Center, Newton, KS, ²Department of Neurology, Centennial Neurosciences, Tristar Centennial, Belmont University, Nashville, TN, USA

Correspondence to: Dr. Mangayakarasi Thandampallayam, University of Kentucky, Lexington, Kentucky, United States. E-mail: mth297@uky.edu

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focused exclusively on GLP-1 RA use in diabetes or obesity without mention of IIH.

Findings from the included literature were synthesized to evaluate the therapeutic potential and challenges of GLP-1 RAs in IIH.

Data Reviewed

The data reviewed for this article included a mix of randomized clinical trials, real-world retrospective studies, and case reports. A key study, the IIH: Pressure trial, was a randomized, placebo-controlled, double-blind trial that provided significant evidence for the direct ICP-lowering effects of exenatide. Other large-scale retrospective cohort studies, particularly those using databases like the TriNetX US Collaborative Network, have provided valuable real-world data on a larger patient population.

These studies consistently found that patients with IIH who were prescribed a GLP-1 RA had significantly lower rates of medication use, fewer symptoms (headaches, visual disturbances), and a reduced need for invasive procedures, such as cerebrospinal fluid (CSF) shunts. Interestingly, some studies noted these improvements occurred even without a significant change in body mass index (BMI) at the 1-year follow-up, suggesting mechanisms of action beyond simple weight loss. Potential mechanisms include a direct effect on the choroid plexus to decrease CSF secretion and a reduction in neuroinflammation.

IIH Pathophysiology

The exact etiology of IIH remains unclear, with multiple theories exploring the role of CSF dynamics. These theories include the contributions of increased cerebral blood flow, impaired CSF resorption, CSF outflow obstruction, and cerebral edema. Additionally, inflammation has been proposed as a potential contributing factor.^[2] The notion of increased cerebral blood or CSF volume was first introduced by Dandy,^[3] and subsequent studies have provided support for the presence of increased cerebral blood volume, particularly through the use of intracarotid tracer injections.^[4]

A strong association exists between obesity and IIH, with a higher body mass index (BMI) correlating with an increased risk of both developing IIH and its recurrence.^[5] It has been proposed that obesity elevates the intra-abdominal pressure, which in turn impedes cerebral venous drainage, leading to an increase in ICP. Furthermore, obesity is often linked to a prothrombotic state and chronic inflammation, both of which may contribute to elevated ICP.^[2]

A strong association exists between IIH and obstructive sleep apnea (OSA), although the role of OSA as an independent risk factor for IIH remains a subject of debate. It has been proposed that OSA may contribute to elevated ICP through mechanisms such as vasodilation and increased cerebral blood flow, both of which are triggered by the hypoxia and hypercarbia associated with apneic events. These physiological changes can potentially lead to alterations in intracranial hemodynamics. Moreover, the obesity commonly observed in individuals with OSA further exacerbates the situation. Obesity is associated with increased intra-abdominal pressure, which can reduce cerebral venous return and impede the efficient drainage of CSF, thus

contributing to an elevation in ICP. The interplay between OSA, obesity, and IIH underscores the complexity of the condition and suggests that managing obesity and addressing OSA may be important components in the treatment and prevention of IIH. However, further research is required to establish a definitive causal relationship between OSA and IIH and to better understand the underlying mechanisms involved.^[6]

Glucagon-Like Peptide-1 Receptor Agonists: Mechanism, Effects on IIH, and Adverse Effects

GLP-1 RAs are a class of incretin-based therapies that enhance the body's physiological response to food intake, promoting glycemic control and weight reduction. These agents mimic the endogenous hormone GLP-1, which is secreted from the gut in response to ingestion of food and stimulates insulin secretion while suppressing glucagon release under hyperglycemic conditions.

Pharmacologically, GLP-1 RAs possess improved pharmacokinetic and pharmacodynamic profiles over native GLP-1, resulting in prolonged receptor activation and enhanced metabolic effects. They bind to GLP-1 receptors distributed across various tissues, including pancreatic islet cells, the brain, kidney, lung, heart, and gastrointestinal tract, exerting both glycemic and extraglycemic benefits.

Mechanistically, GLP-1 RAs act through multiple pathways: (1) enhancing glucose-dependent insulin and amylin secretion from pancreatic β -cells, (2) suppressing glucagon release from α -cells, (3) reducing hepatic glucose production, (4) increasing satiety via central nervous system activation, (5) delaying gastric emptying, and (6) improving insulin sensitivity through weight loss.

A distinguishing feature of GLP-1 RAs is their ability to promote significant weight loss, which is attributed to both central and peripheral effects. Centrally, GLP-1 RAs activate hypothalamic satiety centers and possibly modulate mesolimbic reward pathways, leading to reduced appetite, diminished hunger, and decreased energy intake. Peripherally, delayed gastric emptying contributes to early satiety and lower caloric consumption. Clinical studies and animal models support the notion that GLP-1 RAs influence food-related motivation and behavior through these pathways.^[7]

Evidence from multiple trials underscores the weight-lowering potential of GLP-1 RAs. A large meta-analysis of 47 randomized controlled trials involving 23,244 patients reported an average weight loss of 4.57 kg, a BMI reduction of 2.07 kg/m², and a 4.55 cm decrease in waist circumference. The most pronounced benefits were seen in younger, nondiabetic females with higher baseline BMI and lower HbA1c. Another review focusing on nondiabetic individuals affirmed similar weight-lowering outcomes, highlighting the broader utility of GLP-1 RAs beyond glycemic control.^[8]

Notably, longer-acting agents like semaglutide and dulaglutide provide sustained receptor activation, resulting in greater reductions in fasting glucose and more significant weight loss compared to short-acting formulations. For example, semaglutide demonstrated superior efficacy in reducing body weight versus exenatide extended release and dulaglutide.^[9]

Another notable mechanism of GLP-1 receptor agonists is their potential effect on migraines themselves. A preclinical study demonstrated that activation of GLP-1 receptors by liraglutide significantly suppressed CGRP expression in the trigeminal nucleus caudalis (TNC) of mice with chronic migraine. This reduction in CGRP levels was associated with decreased central sensitization and alleviation of migraine-associated hyperalgesia.^[10] Furthermore, GLP-1 RAs have been shown to alleviate chronic migraine-associated hyperalgesia by inhibiting CGRP, phosphorylated ERK (p-ERK), and c-fos protein levels in the TNC, with a concurrent increase in IL-10 release.^[11]

Glucagon-Like Peptide-1 Receptor Agonists: Potential Mechanistic Effects on CSF Physiology in IIH

GLP-1 RAs may influence CSF dynamics through both indirect and direct mechanisms as illustrated in Flow Charts 1, 2, and 3. The most established pathway is weight reduction, a critical therapeutic target in IIH, as even modest reduction in body weight is associated with reduction in ICP (Dickson *et al.* 2011, Cornell 2020).^[7,12] By suppressing appetite, delaying gastric emptying, and promoting sustained weight loss, GLP-1 RAs indirectly lower the metabolic and hormonal drivers of increased ICP.^[8,12]

Emerging evidence also suggests a weight-independent role of GLP-1 signaling in CSF regulation. GLP-1 receptors are expressed in the choroid plexus and other brain regions

involved in fluid homeostasis. Preclinical studies indicate that GLP-1 receptor activation may reduce CSF secretion, possibly by modulating ion transport and aquaporin channels within the choroid plexus epithelium. This direct effect could complement the pressure-lowering benefits derived from weight loss, providing a dual mechanism of action.^[8]

Taken together, GLP-1 RAs may ameliorate intracranial hypertension through synergistic mechanisms: systemic metabolic effects that drive weight loss and potential central actions that reduce CSF production. These pathways highlight their therapeutic promise in IIH, warranting further mechanistic and clinical investigation. Table 1 provides a list of FDA-approved GLP-1 receptor agonists.

Glucagon-Like Peptide-1 Receptor Agonists: Adverse Effects

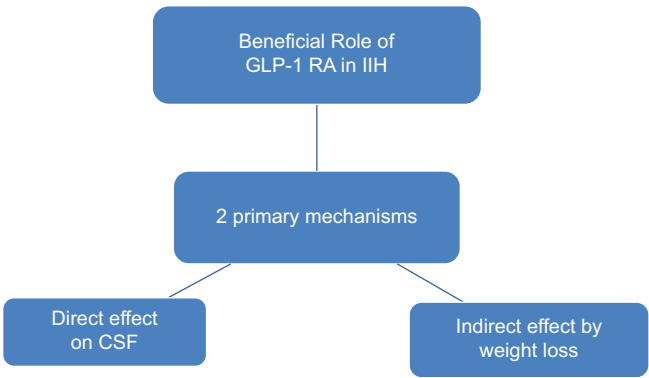
GLP-1 receptor agonists are commonly associated with gastrointestinal (GI) adverse effects, which are often most pronounced at the beginning of treatment and tend to diminish over time. Among GI symptoms, nausea and diarrhea are very common ($\geq 1/10$), while vomiting, constipation, abdominal pain, and dyspepsia are relatively common ($\geq 1/100$ to $< 1/10$) (Filippatos *et al.*, 2014).^[13] Nausea is the most frequently reported side effect, affecting up to 50% of patients, and usually appears in mild to moderate, dose-dependent episodes.^[13]

Concerns have also been raised regarding the risk of pancreatitis. Preclinical studies show that drugs like exenatide and sitagliptin can induce pancreatic inflammation, whereas liraglutide shows only a modest association.^[14]

Injection site reactions such as rash, erythema, and itching are common. While GLP-1 receptor agonists used with metformin do not increase hypoglycemia risk, the combination with sulfonylureas or insulin can increase hypoglycemia incidence.^[15]

Exenatide has also been linked to acute kidney injury (AKI), primarily due to dehydration from GI symptoms.^[16]

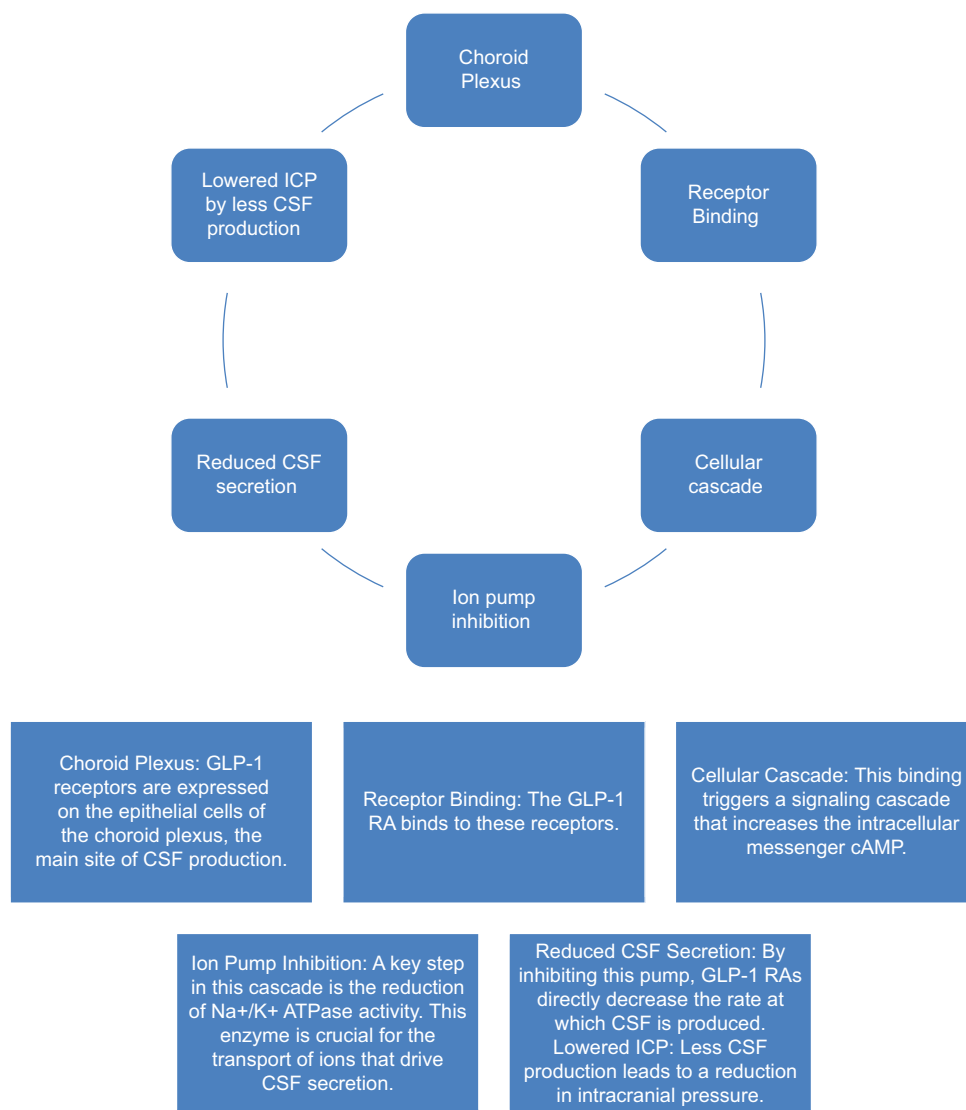
Emerging evidence also suggests a possible link between GLP-1 receptor agonists and non-arteritic anterior ischemic optic neuropathy (NAION). A 2023 case series from Massachusetts Eye and Ear by Hathway *et al.*^[17] reported sudden vision loss in patients shortly after initiating semaglutide therapy.



Flow Chart 1: Proposed mechanisms of GLP-1 RA in IIH

Table 1: Pharmacological characteristics of FDA-approved GLP-1 receptor agonists

GLP-1 Agonist	Trade Name (USA)	Company	FDA Approval Year	Initial Dose	Regular Dose	T_{max}	Half-Life (t_½)	Elimination
Exenatide	Byetta®	Amylin Pharmaceuticals	2005	5 mcg BID	5 or 10 mcg BID	2.1 h	2.4 h	Renal/proteolysis
Lixisenatide	Adlyxin®	Sanofi-Aventis	2016	10 mcg	20 mcg	1–3.5 h	3 h	Renal/proteolysis
Liraglutide	Victoza®	Novo Nordisk	2010	0.6 mg	1.2 or 1.8 mg	8–12 h	13 h	No specific organ
Exenatide ER	Bydureon®	AstraZeneca	2012	2.0 mg	2.0 mg	Peaks at 2 & 6–7 wks	5 days	Renal/proteolysis
Dulaglutide	Trulicity®	Eli Lilly	2014	0.75 mg	0.75 or 1.5 mg	24–72 h (median 48 h)	5 days	Endogenous metabolism
Semaglutide	Ozempic®	Novo Nordisk	2017	0.25 mg	0.5 or 1.0 mg	1–3 days	7 days	Urine and feces
Tirzepatide	Mounjaro®	Eli Lilly	2022	2.5 mg	5, 10, or 15 mg	~1 week	~5 days	Urine and feces



Flow Chart 2: Proposed mechanisms of GLP-1 RA in IIH: Direct effect on CSF production

Last, rodent studies have associated GLP-1 RAs with medullary thyroid carcinoma, though these findings have not been confirmed in primate models, suggesting species-specific effects.^[18]

Glucagon-Like Peptide-1 Receptor Agonists: Limitations and Challenges

The use of GLP-1 RAs for weight loss in patients with IIH presents several challenges and limitations. A major concern is the limited availability of IIH-specific clinical data regarding GLP-1 RAs. Most existing evidence is extrapolated from studies in populations with obesity and/or type 2 diabetes, leaving uncertainty about the true efficacy and safety of these agents in the IIH population. Although weight reduction is known to improve IIH symptoms, it remains unclear whether GLP-1 RAs exert any direct effect on ICP or CSF dynamics.

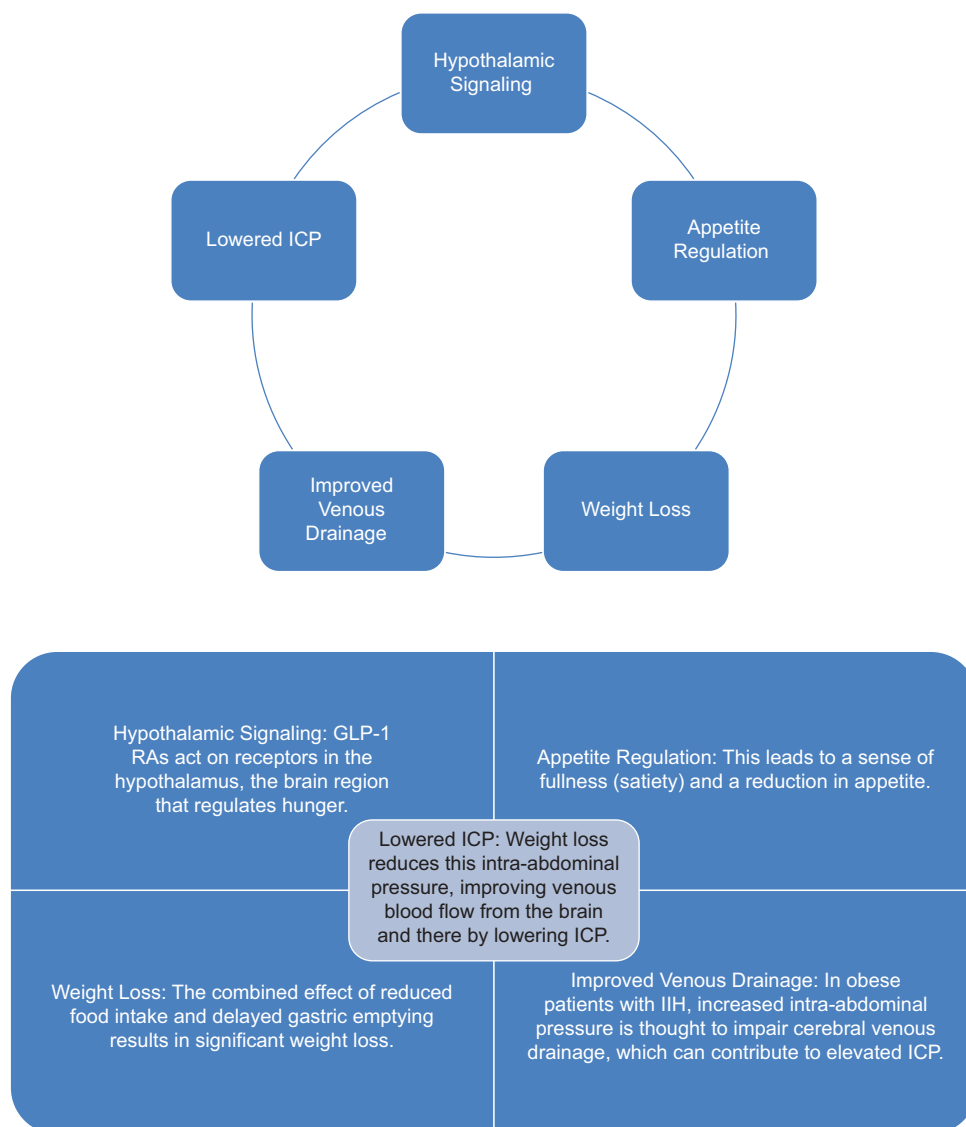
Additionally, the high cost and limited insurance coverage for off-label use may restrict access to these medications. Side effects, including nausea, GI discomfort, and the risk of

pancreatitis, may further impact tolerability and long-term adherence.

Since sustained symptom relief often depends on continued weight management, patients may require prolonged therapy, which raises concerns about long-term safety, affordability, and commitment. Finally, it is important to emphasize that pharmacologic therapy should be viewed as part of a comprehensive, multidisciplinary approach to IIH management, rather than as a standalone treatment, to ensure optimal patient outcomes. Using GLP-1 RAs as a main therapy could potentially detract from comprehensive lifestyle and medical interventions that are crucial for management of IIH.

Clinical Evidence and Ongoing Trials of GLP-1 RAs in IIH

Botfield *et al.* showed that GLP-1 R was expressed in human and rodent choroid plexus. His study showed that GLP-1R agonists reduced ICP in the rat model of hydrocephalus indicating potential benefits in patients with IIH in addition to weight loss.^[19]



Flow Chart 3: Proposed Mechanisms of GLP-1RA in IIH: Indirect Effect by Weight loss

The role of Exenatide in IIH was studied in a Phase 2 trial, that is, the PRESSURE trial, which is a double-blinded placebo-controlled study where patients were enrolled to receive either subcutaneous exenatide or placebo. The participants were women with active IIH with a documented ICP > 25 cm of water and papilledema. The ICP was measured by telemetric ICP catheters. The outcomes were noted at 2.5 h, 24 h, and 12 weeks. A total of 15 participants completed the study (Mitchell *et al.*, 2023).^[1] Exenatide was found to have caused a statistically significant reduction in ICP in the immediate time period and continued to provide sustained reduction at the 3-month mark. The pressure reduction noted at the predetermined 2.5 h was -5.7 ± 2.9 cm CSF ($P = 0.048$), that at 24 h was -6.4 ± 2.9 cm CSF ($P = 0.030$), and that at 12 weeks was -5.6 ± 3.0 cm CSF ($P = 0.058$). The most common adverse effect was nausea, seen in about 85% of the participants. Positive results from the PRESSURE trial paved the way for 2 Phase 3 trials, IIH EVOLVE and IIH ADVANCE study (Invex therapeutics). However, the study was determined to be

nonviable and was terminated by the Invex board. Table 2 provides a summary of various studies of GLP-1 RAs in IIH.

Kravetz *et al.*^[20] conducted a retrospective cohort study on 79 patients with IIH comparing two groups of patients, one that were treated solely with Acetazolamide alone (68) and the other group treated with a GLP-1 RA in addition to Acetazolamide (11). The study did not show any significant differences in the following parameters including the papilledema grade, best corrected visual acuity, visual field, or OCT RNFL thickness improvements over a 12-month period. 7/11 patients on the combined therapy completed 1 full year of follow-up. Data showed a positive reduction in BMI and thus effective reduction in the acetazolamide dose, however, without changing the clinical outcome.^[20]

Conclusion

Evidence from multiple clinical trials underscores the promising role of GLP-1 RAs in the management of IIH as

Table 2: Comparative table on various studies

Study	Design	Population	Duration	Outcome	Limitations
Mitchell <i>et al.</i> (IIH:Pressure Trial) (2023)	Randomized, placebo-controlled, double-blind trial	16 women with IIH (age 18-60) and a mean BMI of 38.1 kg/m ²	12 weeks	Primary: Significant reduction in ICP at 2.5 hours, 24 hours, and 12 weeks. Secondary: No serious safety signals.	Small sample size, short duration, only included women. The study was not powered for clinical outcomes like headache or vision.
TriNetX Retrospective Cohort Study (2025)	Retrospective cohort study with propensity score matching	44,373 patients with IIH (after matching, 555 GLP-1 RA users vs. 555 non-users)	1 year	Primary: Lower medication use, reduced headaches, and fewer visual disturbances or papilledema. Secondary: Lower need for procedures (shunts) and reduced mortality.	Retrospective design, which is prone to confounding factors. Causal relationships cannot be definitively established.
Azzam <i>et al.</i> (2024) ^[22]	Retrospective propensity score-matched study	Patients with IIH treated with tirzepatide vs. a control group	24 months	Primary: Significant reduction in papilledema, visual disturbance, and headache risk. Secondary: Significant BMI reduction.	Retrospective, non-peer-reviewed preprint. Findings need to be confirmed by prospective studies.
TriNetX Retrospective Cohort Study on Clinical/ Neurosurgical Outcomes (2025)	Propensity-matched multi-institutional cohort study	~5,000 patients with IIH (GLP-1 RA users matched to control group)	6 months and 1 year	Primary: Significant reduction in BMI, headaches, visual deficits, and cognitive deficits. Secondary: Lower odds of acetazolamide use and shunt placement.	Retrospective nature, which limits definitive conclusions about causality.
Krajnc <i>et al.</i> (2023)	Open-label, single-center, case-control pilot study	39 patients with IIH (13 on GLP-1 RA, 26 on usual care) with BMI ge 30 kg/m ²	6 months	Primary: Significantly higher mean weight loss in GLP-1 RA group. Secondary: Greater reduction in monthly headache days.	Small sample size, open-label design, potential for selection bias.

outlined in Table 2. The therapeutic potential of these agents lies in their unique synergistic mechanism: They drive systemic metabolic effects that facilitate significant weight loss while simultaneously exerting central actions that may directly reduce CSF production. Unlike current treatments like acetazolamide, which act as a diuretic, GLP-1 RAs appear to directly influence the production of CSF in the choroid plexus, a primary site of GLP-1 receptor expression in the brain in addition to weight loss. Studies have shown that GLP-1 RAs are associated with significant reductions in headache frequency, severity, and visual deficits. They also lower the need for procedural interventions and may even be associated with reduced mortality. Since IIH is so strongly linked to obesity, GLP-1 RAs offer a dual-purpose treatment by addressing both the symptoms of IIH and its primary risk factor.

While encouraging, much of the current data come from retrospective cohort studies; more randomized, prospective clinical trials are needed to confirm the efficacy, safety, and optimal dosing of GLP-1 RAs specifically for IIH. The studies are also limited by small sample sizes, short follow-up durations, retrospective designs, and some studies included only women. The long-term effects of GLP-1 RA use in patients with IIH are still under investigation. Potential risks, such as pancreatitis and thyroid tumors, are a concern, although these are generally considered rare. Case reports have shown that abrupt cessation of GLP-1 RAs can lead to rapid weight regain and a recurrence or worsening of IIH symptoms.

Consequently, there is a need for large-scale, prospective clinical trials with extended observational periods. Such

research is essential to fully elucidate the role of GLP-1 RAs in the IIH treatment paradigm and to determine their long-term impact on preventing permanent visual loss.^[21]

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First author:

Mangayakarasi Thandampallayam

Assistant Professor,
University of Kentucky,
Lexington, Kentucky, USA